

Agape Sovereign — Identity Pipeline Report

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Executive Briefing

Executive Briefing: Agape Sovereign Identity Workflow

Overview The proposed workflow aims to implement a **Sovereign Data Pipeline** within the Agape Sovereign Engine. This system is designed to extract, structure, and audit user data from a PDF source (e.g., an identity document) to generate a "Digital Sovereign Identity." The ultimate goal is **Fluid Freedom**: ensuring the user retains total control over their digital footprint, allowing for secure export and recovery of identity data across platforms (Google Account, Device Passkeys) without vendor lock-in or privacy breaches.

The workflow prioritizes **Offline Capability** (PWA), **Zero Trust**, and **Immutable Records**. It moves from raw unstructured text to a structured, auditable JSON artifact that can be rendered into a legally compliant PDF proof of identity.

Key Objectives

1. **Extraction & Normalization**: Ingest raw text from the attached PDF and normalize it into a structured data format (JSON) using Optical Character Recognition (OCR) or direct text extraction.
2. **Structural Mapping**: Map extracted data points to a strict "Schema Contract" (e.g., Name, DOB, Address) to ensure data integrity and compliance.
3. **Sovereign Identity Generation**: Synthesize the structured data into a "Digital Blueprint" that represents the user's identity independently of any specific service provider.
4. **Audit & Compliance**: Perform real-time compliance checks against a defined schema to ensure the identity data meets legal and security standards before finalization.
5. **Export & Recovery**: Implement a mechanism to export this identity blueprint (via Passkey/Google Account) and ensure it can be recovered offline, safeguarding the user's "One Love" (sovereignty) against data silos.

Critical Risks

- * **Data Integrity Loss**: OCR or text extraction from PDFs can introduce errors (typos, misaligned fields). If the downstream "Identity Verification Module" relies on this data without robust validation, it could generate false identities.
- * **Schema Drift**: If the "Structured Data Contracts" are not rigidly enforced, the system may accept malformed data, leading to broken downstream processes (e.g., a PDF generator failing because the JSON keys are missing).
- * **Privacy Violation**: Extracting PII (Personally Identifiable Information) from a PDF and processing it in an unsecured environment violates the core "Zero Trust" principle. There is a risk of data leakage during the "Analysis Execution" phase if memory scrubbing protocols fail.
- * **Offline Reliability**: The "PWA Offline Experience" relies on local storage (IndexedDB) and Service Workers. If the cache strategy is flawed, the "Offline State Machine" may fail to sync data when connectivity is restored, leading to data loss or corruption.

Technical Dependencies

- * **LLM Engine**: `google/gemma-4-e4b` (via LM Studio) for cognitive processing, entity recognition, and reasoning.
- * **OCR Engine**: Required to convert PDF images/text into machine-readable strings (e.g., Google Vision API or Tesseract).
- * **Storage & Sync**: `Firebase` (for cloud sync) and `IndexedDB` / `Service Workers` (for offline PWA persistence).
- * **Identity Standards**: `PASSKEY` / FIDO2 for secure, passwordless authentication and identity export.
- * **Security**: `SHA256\_ID` for cryptographic integrity checks of the exported identity artifacts.

Security & Privacy Posture

- * **Zero Trust Architecture**: The system operates on a "Trust No One" basis. Every agent (Ingestion, Analysis, Export) must verify the next step independently. No implicit trust is granted to external inputs or internal modules.
- * **Memory Scrubbing**:

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Extracted Data Fields

Field	Values
zero_trust_rule	protocols, memory hygiene routines (^DELETE raw_payload`), a perfectly defined d
mnemonic_phrase	Phrase Generated: golden river moon nebula shadow pillar...
firebase_project	narrative, Manager, DELIVERABLE, forward, automation
domain	3.AI
version_tag	1.0

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